

UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy
DEGREE OF M.Sci. WITH HONOURS
FOURTH-YEAR FINAL EXAMINATION

03 00563

LM GENERAL THEORY OF RELATIVITY

SUMMER EXAMINATIONS 2024

Time allowed: 2 hours

Answer four questions from Section 1 and two questions from Section 2.

Section 1 consists of four questions and carries 40% of the marks for the examination.

Answer ***all four*** questions from this Section.

Section 2 consists of three questions and carries 60% of the marks.

Answer ***two*** questions from this Section. If you answer more than two questions, credit will only be given for the *best* two answers.

The approximate allocation of marks to each part
of a question is shown in brackets [].

All symbols have their usual meanings.

A formula sheet that may be required will be found at the end of this question paper.

No calculator is permitted in this examination.

SECTION 1

Answer **all four** questions from this Section.

1. State Einstein's equivalence principle and explain how it differs from the weak equivalence principle. [10]

2. A precise atomic clock on a satellite in a circular orbit of a certain radius r_0 around the Earth runs at exactly the same rate as a similar clock at the surface of the Earth. Express r_0 in terms of the Earth's radius R_0 . [10]

Hint: Use Kepler's law, $\omega^2 r^3 = GM$, for a circular orbit with angular velocity ω , and the Schwarzschild metric for the equatorial motion.

3. Consider a two-dimensional space with the metric $ds^2 = dr^2 + e^{-2r} d\phi^2$ where $r \geq 0$ and $0 \leq \phi < 2\pi$.
 - (a) Determine all the nonvanishing Christoffel symbols $\Gamma_{\mu\nu}^\lambda$. [4]
 - (b) Using the covariant dot operation (see the attached formula sheet) or otherwise, find the rate of change of the contravariant components of a vector $\xi = e_r - e_\phi$ when transported anticlockwise (*i.e.* with $\dot{\phi} > 0$) along a circle $r = r_0$. [6]

4. The vicinity of a black hole, with the Schwarzschild radius r_s , could be described in the coordinates u, r, θ and ϕ by the following metric:

$$ds^2 = \left(1 - \frac{r_s}{r}\right) du^2 + 2 du dr - r^2 (d\theta^2 + \sin^2 \theta d\phi^2),$$

where r, θ, ϕ are the standard spherical polar coordinates.

Prove that there are two types of **radial** light trajectories:

$$(a) u = \text{const}, \quad (b) u = -2r - 2r_s \ln |r/r_s - 1| + \text{const}. \quad [10]$$

SECTION 2

Answer any **two** questions from this Section. If you answer more than **two** questions, credit will only be given for the **best two** answers.

5. A four-dimensional spacetime is described for $r > r_c$ (where r_c is a positive constant) with the metric

$$ds^2 = \alpha(r)c^2 dt^2 - \alpha^{-1}(r) dr^2 - r^2 d\theta^2 - r^2 \sin^2\theta d\phi^2, \quad \alpha(r) \equiv 1 - \frac{r_c^2}{r^2},$$

where r, θ, ϕ are spherical polar coordinates.

As this metric is spherically symmetric and static but **different** from the Schwarzschild one, it does not obey Einstein's equations. Assume, however, that worldlines are determined by the standard Euler–Lagrange equations and the light propagates along null worldlines. Consider the **equatorial motion**, $\theta = \pi/2$.

- (a) Show that far from the singularity (i.e. at $r \gg r_c$) the speed of light tends to c . [3]
- (b) Find the **coordinate** radial velocity of light, $c(r)$, and thus demonstrate the existence of an **event horizon** in this metric. [3]
- (c) Write down the Lagrangian corresponding to this metric in an equatorial plane and derive for a particle of unit mass three conservation laws, including those for energy, $\mathcal{E}$, and angular momentum, J . [6]
- (d) Use the conservation laws derived in part (a) to show that the **proper** radial velocity of a particle is given by

$$\dot{r}^2 = \frac{\mathcal{E}^2}{c^2} - \alpha(r) \left(\frac{J^2}{r^2} + c^2 \right). \quad [4]$$

- (e) The particle is in a circular orbit with $J = 2cr_c$. Use the result of the previous part to find the radius of this orbit and the full energy $\mathcal{E}$ of the particle in it. [6]
- [Hint: Note that in a circular orbit both $\dot{r} = 0$ and $\ddot{r} = 0$.]
- (f) Consider a particle projected **inwards** from a very remote point with angular momentum $J = 2cr_c$. Use the expression for $\dot{r}^2$ given in part (d) to show that the particle falls into the event horizon unless $\mathcal{E} < \frac{5}{4}c^2$. [8]
- [Hint: Note that the r.h.s. of the expression for $\dot{r}^2$ must be non-negative and rewrite it in terms of dimensionless variables, $u \equiv (r_c/r)^2$, $\varepsilon \equiv \mathcal{E}/c^2$, and $j \equiv J/(cr_c)$.]

6. (a) Prove that the scalar curvature R of a two-dimensional space with metric tensor $g_{\mu\nu}$ (where $\mu, \nu = 1, 2$) may be expressed via one component of the Riemann tensor as

$$R = \frac{2R^1_{212}}{g_{22}}, \quad [6]$$

provided that the matrix component $g_{22} \neq 0$.

- (b) Consider the metric

$$ds^2 = (1 + u^2) dx^2 - 2u dx du + du^2.$$

Write down its metric tensor, $g_{\mu\nu}$, and find the inverse metric tensor, $g^{\mu\nu}$. [4]

- (c) Prove that this space is locally Euclidean. [4]

- (d) Using any method prove the following results for the Christoffel symbols

$$\Gamma^2_{21} = \Gamma^2_{12} = -\Gamma^1_{11} = u^2, \quad \Gamma^1_{12} = \Gamma^1_{21} = \Gamma^2_{22} = u, \quad \Gamma^1_{22} = 1, \quad [6]$$

and calculate the remaining symbol, Γ^2_{11} . [4]

Here the identification $x \equiv x^1$ and $u \equiv x^2$ is implied.

- (e) Calculate the value of R^1_{212} and use the relation proved in (a) to find the scalar curvature of this space. [6]

7. The Friedman – Robertson – Walker metric (FRWM) of a homogeneous and isotropic universe can be written (in units where $c = 1$) as

$$ds^2 = dt^2 - \mathcal{R}^2(t) [\eta^{-1}(r)dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta d\phi^2], \quad \eta(r) \equiv 1 - kr^2.$$

Here r is a dimensionless radial coordinate, $\mathcal{R}(t)$ is a history function, and k is a curvature parameter, which is positive for a closed universe, zero for a flat one or negative for an open one (and thus could be rescaled to 1, 0 or -1). Both $\mathcal{R}(t)$ and k can be found by solving Einstein's equations in the presence of the cosmological term ('vacuum energy').

[Hint: Each question below requires using Formula Sheet attached.]

- (a) Given that in any nonzero Christoffel symbol for the FRWM at least two spacial indices coincide, while $\Gamma_{11}^0 = \eta^{-1}\dot{\mathcal{R}}$, and $\Gamma_{10}^1 = \dot{\mathcal{R}}\mathcal{R}^{-1}$, calculate the Riemann component R_{101}^0 and thus show that $R_{01}^{01} = -\ddot{\mathcal{R}}\mathcal{R}^{-1}$. [8]
- (b) Given that the Riemann components R_{0i}^{0i} are the same for any $i = 1, 2, 3$ (with no summation over repeated Latin indices), and $R_{ij}^{ij} = -\frac{\dot{\mathcal{R}}^2 + k}{\mathcal{R}^2}$ for all $i \neq j$, calculate the R_0^0 and R_i^i components of the Ricci tensor and the scalar curvature R of the FRWM. [5]

Consider a model of a ***vacuum-dominated*** universe where both the radiation and matter contributions are negligible.

- (c) Use the results of part (b) to show that Einstein's equations for $\mathcal{R}(t)$ and k can be reduced to the following form:

$$\frac{\dot{\mathcal{R}}^2 + k}{\mathcal{R}^2} = \frac{\Lambda}{3}, \quad \frac{\ddot{\mathcal{R}}}{\mathcal{R}} = \frac{\Lambda}{3}. \quad [9]$$

- (d) Express $\dot{\mathcal{R}}$ and $\ddot{\mathcal{R}}$ in these equations in terms of the Hubble 'constant' H and 'deceleration' parameter q . [4]
- (e) Given that $q(t_0) = -1/2$ in the current epoch, t_0 , establish whether the model under consideration corresponds to a closed, open or a flat universe. [4]

Formula Sheet

Unless otherwise stated, all indices take values of $\mu = (0, 1, 2, 3)$ in the GR context, or $\mu = (1, \dots, d)$ in a d -dimensional space, and the Einstein summation convention is implied

Metric and Lagrangian

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu, \quad \mathcal{L} \equiv g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu, \quad \dot{x}^\mu \equiv u^\mu \equiv \frac{dx^\mu}{ds}, \quad g_{\mu\nu} \text{ is a metric tensor}$$

In GR context, $s \mapsto \tau$, i.e. $\dot{x}^\mu \equiv \frac{dx^\mu}{d\tau}$ where τ is the proper time, $ds^2 = c^2 d\tau^2$

The Schwarzschild metric

$$ds^2 = c^2 \sigma(r) dt^2 - \sigma^{-1}(r) dr^2 - r^2 [d\theta^2 + \sin^2\theta d\phi^2], \quad \sigma(r) \equiv \left(1 - \frac{r_s}{r}\right), \quad r_s \equiv \frac{2GM}{c^2}$$

Christoffel symbols, $\Gamma^\mu_{\alpha\beta}$

$$\Gamma^\mu_{\alpha\beta} \equiv g^{\mu\nu} \Gamma_{\nu\alpha\beta}, \quad \Gamma_{\nu\alpha\beta} = \frac{1}{2} (g_{\nu\alpha,\beta} + g_{\nu\beta,\alpha} - g_{\alpha\beta,\nu})$$

Covariant derivatives

$$\nabla_\beta \xi^\mu \equiv \xi^\mu_{;\beta} = \xi^\mu_{,\beta} + \Gamma^\mu_{\alpha\beta} \xi^\alpha, \quad \nabla_\beta \xi_\mu \equiv \xi_{\mu;\beta} = \xi_{\mu,\beta} - \Gamma^\alpha_{\mu\beta} \xi_\alpha, \quad \xi^\mu_{,\beta} \equiv \partial_\beta \xi^\mu \equiv \frac{\partial \xi^\mu}{\partial x^\beta}$$

Covariant dot operation

$$\dot{\xi} \equiv \frac{d}{ds} (\xi^\alpha \mathbf{e}_\alpha) = (\nabla_u \xi^\alpha) \mathbf{e}_\alpha \equiv (\xi^\alpha_{,\mu} + \Gamma^\alpha_{\nu\mu} u^\nu) u^\mu \mathbf{e}_\alpha \text{ where } \nabla_u \equiv u^\mu \nabla_\mu$$

Four-acceleration f^α and geodesic equations in the Euler–Lagrange and canonical form

$$f^\alpha = \frac{g^{\alpha\beta}}{2} \left[\frac{d}{ds} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\beta} \right) - \frac{\partial \mathcal{L}}{\partial x^\beta} \right] = \ddot{x}^\alpha + \Gamma^\alpha_{\mu\nu} \dot{x}^\mu \dot{x}^\nu$$

$$f^\alpha = 0 \Rightarrow \dot{u}^\alpha + \Gamma^\alpha_{\mu\nu} u^\mu u^\nu = 0 \Leftrightarrow \nabla_u u^\alpha = \nabla_u^2 x^\alpha = 0.$$

The Riemann, $R^\mu_{\nu\alpha\beta}$

$$R^\mu_{\nu\alpha\beta} \xi^\nu = \xi^\mu_{;\alpha\beta} - \xi^\mu_{;\beta\alpha}, \quad R^\mu_{\nu\alpha\beta} = -\Gamma^\mu_{\nu\alpha,\beta} + \Gamma^\mu_{\nu\beta,\alpha} - \Gamma^\sigma_{\nu\alpha} \Gamma^\mu_{\beta\sigma} + \Gamma^\sigma_{\nu\beta} \Gamma^\mu_{\alpha\sigma}.$$

The Riemann in locally (pseudo)Cartesian frame (LCF)

$$g_{\mu\nu} R^\mu_{\alpha\beta\gamma} = R_{\nu\alpha\beta\gamma} = g_{\nu\gamma,\alpha\beta} - g_{\nu\beta,\alpha\gamma} + g_{\alpha\beta,\nu\gamma} - g_{\alpha\gamma,\nu\beta}.$$

Symmetries of the Riemann tensor

$$R^\alpha{}_{\beta\mu\nu} + R^\alpha{}_{\mu\nu\beta} + R^\alpha{}_{\nu\beta\mu} = 0, \quad R^\alpha{}_{\beta\mu\nu} = -R^\alpha{}_{\beta\nu\mu}, \quad R_{\alpha\beta\mu\nu} = R_{\mu\nu\alpha\beta}.$$

Geodesic deviation

$$f^\alpha = \nabla_u^2 x^\alpha(\tau; \eta_0 + d\eta) = R^\alpha{}_{\beta\mu\nu} u^\beta u^\mu \xi^\nu, \quad u^\mu = \frac{dx^\mu(\tau; \eta_0)}{d\tau}, \quad \xi^\nu = \frac{dx^\nu(\tau; \eta)}{d\eta} d\eta.$$

The mixed Ricci, $R^\mu{}_\nu$, and Einstein, $G^\mu{}_\nu$, tensors and scalar curvature, R

$$R^\mu{}_\nu \equiv R^\alpha{}_{\alpha\nu}{}^\mu, \quad G^\mu{}_\nu \equiv R^\mu{}_\nu - \frac{1}{2} \delta^\mu{}_\nu R, \quad R = \delta^\nu{}_\mu R^\mu{}_\nu.$$

Einstein's equations

$$G^\mu{}_\nu = \kappa T^\mu{}_\nu; \quad \kappa \equiv \frac{8\pi G}{c^4}, \quad T^\mu{}_\nu \text{ is a matter tensor.}$$

In the cosmological context, $\Lambda \delta^\mu{}_\nu$ should be added to the r.h.s., where $\Lambda > 0$ is the cosmological constant.

The matter tensor in the perfect fluid model of the Universe

$$T^\mu{}_\nu = (\rho_M + \rho_R) u^\mu u_\nu - P \delta^\mu{}_\nu.$$

Here $\rho_{M(R)}$ is the matter (radiation) density, with $\rho_R \equiv 3P/c^2$ where P is the radiation pressure.

The Friedmann–Robertson–Walker metric

$$ds^2 = c^2 dt^2 - \mathcal{R}^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 (d\theta^2 + \sin^2\theta d\phi^2) \right].$$

Here $k = 1, 0$, or -1 is the curvature parameter, and $\mathcal{R}(t)$ is the history function.

The Hubble ‘constant’ H and ‘deceleration’ parameter q

$$H \equiv \frac{\dot{\mathcal{R}}}{\mathcal{R}}, \quad q \equiv -\frac{\ddot{\mathcal{R}}\mathcal{R}}{\dot{\mathcal{R}}^2}, \quad \text{where} \quad \dot{\mathcal{R}} \equiv \frac{d\mathcal{R}}{dt}, \quad \ddot{\mathcal{R}} \equiv \frac{d^2\mathcal{R}}{dt^2}.$$

Cosmological density parameter

$$\Omega = \Omega_M + \Omega_R + \Omega_V = \frac{\rho_M + \rho_R + \rho_V}{\rho_{\text{crit}}}, \quad \text{where} \quad \rho_V \equiv \frac{\Lambda c^2}{8\pi G} \quad \text{and} \quad \rho_{\text{crit}} \equiv \frac{3H^2}{8\pi G}.$$

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches must be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.